

A simulation study investigating a novel method for emotion transfer between virtual humans

Samad Roohi*

Department of Computer Science and Information Technology
La Trobe University

Richard Skarbez†

Department of Computer Science and Information Technology
La Trobe University

ABSTRACT

Affective – that is, emotionally aware – interactions with virtual humans contribute to improved user experience in both immersive and non-immersive virtual worlds. Some important open problems in this area are how to generate plausible moment-to-moment emotions in virtual humans, how to enable and model the transfer of emotions between multiple virtual humans, and how these emotions change and are changed by the environment. Motivated by these open questions, this paper presents a simulation study for the computational models of emotion and its shift and transference between characters.

Keywords: Virtual humans, emotion transfer, appraisal, inference.

1 INTRODUCTION

Advances in the design and implementation of virtual humans have encouraged researchers to incorporate mental states such as emotions in their designs. Research into such socio-affective virtual humans seeks to address several key questions such as: (1) how to generate plausible moment-to-moment emotional states in virtual humans, (2) how to enable and model the transfer of emotions between multiple virtual humans, and (3) how these emotions change and are changed by the environment in which these characters are situated.

Appraisal-based computational models of emotion have been widely employed in the problem of emotion generation. However, their utilization is impeded by the lack of operational insights. This paper uses Scherer's viewpoint which defines emotion as a component process model (CPM) arising from an appraisal of the situation [5]. CPM enables the production of emotional states through a cognitive evaluation of the situation. Thus far, various methods have been proposed to model emotion transfer in a shared virtual/real space. The main thread of research in the literature has suggested modeling emotional contagion through traditional information-spreading models [3]. Previous research has treated emotion transfer as a one-off process in which an emoter conveys part of their emotion to an observer. However, in psychology, emotion transfer is considered a cumulative function of emotion conveyed by the emoter and social appraisal by the observer [4]. Despite limited viewpoints on emotion transfer, previous methods produced valuable insights into safety evacuation and product recommendations. This paper reports on a simulation study to investigate a psychologically valid computational model for emotion generation and its transference between virtual humans.

* s.roohi@latrobe.edu.au

† r.skarbez@latrobe.edu.au

To this end, we developed a dynamic need system to produce stimuli for emotions. The goal-oriented action planning (GOAP) and utility-based AI algorithms have been used to manipulate agents' needs and the CPM model is used to generate subjective emotions by evaluating agents' need and their satisfaction. Then, a model of contagion and social appraisal is used to develop a psychologically valid model for emotion convey between multiple virtual humans.

2 THEORY AND IMPLEMENTATION

To provide a context for testing our proposed method and establish a social milieu for transferring emotion among virtual agents, an interactive virtual environment called "Town Square" is designed and implemented (see Figure 1).

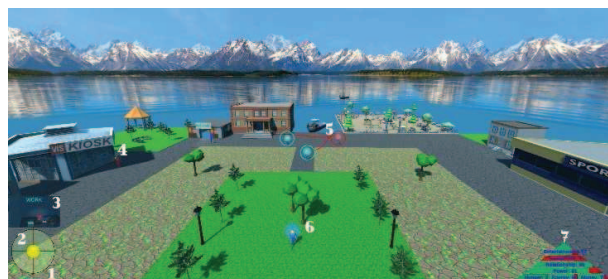


Figure 1: Town Square: (1) the emoter's emotion, (2) the emoter's core affect, (3) the current event, (4) the main emoter, (5) the transfer of emotion (6) an observer, (7) the emoter's needs.

In Town Square, agents are either "emoters" or "observers". Emoters are agents that diffuse emotion and observers are typical agents that can receive emotion if they couple with an emoter. For emoters, emotion is computed using the CPM model. In summary, three stages are performed to produce emotion for an emoter: (1) appraisal of the situation, (2) extracting emotion from the appraisal vector, and (3) expressing dominant emotion. For (1), the system evaluates a typical situation (agent's current needs and world state) on sixteen appraisal dimensions, corresponding to the major categories of Scherer's appraisal theory. This procedure yields a subjective evaluation of a situation. To choose the output emotion, we apply the mapping table proposed by Scherer in [5] on the appraisal dimensions and then use the correlation table introduced in [6] to translate the resulting index into the dimensional "core affect" space.

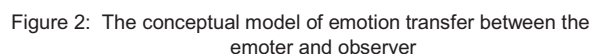
We suggest a collaborative model for emotion transfer, in which observers infer the emotional states of emoters from two sources: (1) emotion contagion: the share of transferred emotion expressed/transmitted directly by emoter, and (2) social appraisal: the part of emotion inferred by the observer (concerning the current

Affective contagion is the part of emotion transferred arbitrarily by the emoter to the observer(s) via, for example, facial expression. Our proposed formulation for affective contagion is inspired by the equation for heat transfer by conduction. The adapted formulation for affective contagion is as follows:

The intensity of infection by *affective contagion* is determined by an agent’s personality. Personality traits impact the emotion transfer between people. We employ the OCEAN/Big Five [1] model to bring personality to virtual humans. Among those traits, extroversion and neuroticism determine the magnitude of emotion transmission from emoter to observer. Also, personality traits impact empathy, which determines the ability of observers to receive emotion from emoters. In (2), X and N denotes weights of extroversion and neuroticism, respectively. In our simulation, all the agents are generated with random personality factors.

In our approach to emotion transfer, a naïve Bayes classifier is used to put a figure to the observer’s inference of emoter’s situation. To this end, we trained a naïve Bayes classifier on the dataset that we captured from the simulation of our CPM-based computational model of emotion. The observer updates their emotion using the following equation:

where A_c is the share of emotion obtained by *affective contagion*, C_c is the *cognitive contagion* inferred by the observer considering its own information about the emoter's situation, and ε_o and ε_1 are weight factors. Figure 2 illustrates the conceptual model of emotion transfer between emoter and observer.


$$v_t = v_{t-1} - e^{\alpha t}, \quad a_t = a_{t-1} - e^{\alpha t}. \quad (4)$$

2.1 Results and Conclusion

Figure 3 depicts the dynamics of emotional transfer between the emoter and 5 observers. During 400 seconds of simulation, the emoter took on 15 emotional states (numbers 1 to 15), denoted by emojis on their start point. Emojis are assigned to each index based on their common use designated in <https://emojipedia.org/>.



Although the results successfully demonstrated the subjectiveness and responsiveness of the proposed method, it has certain limitations. First, to simplify implementation, it relies on some modifications on the theoretical model. In addition, the model's parameters are indicated using heuristic approaches. In terms of future work, a subjective study will be conducted to investigate the effectiveness of the proposed method in real-world scenarios.

- [1] F. Durupinar, N. Pelechano, J. Allbeck, U. Gbay, and N. I. Badler, "How the Ocean Personality Model Affects the Perception of Crowds," *IEEE Computer Graphics and Applications*, vol. 31, no. 3, pp. 22–31, May 2011.
- [2] E. Fernndez-Rodicio, M. Maroto-Gmez, . Castro-Gonzlez, M. Malfaz, and M. . Salichs, "Emotion and Mood Blending in Embodied Artificial Agents: Expressing Affective States in the Mini Social Robot," *Int J of Soc Robotics*, Sep. 2022.
- [3] X. Nan, Z. Zehong, and P. Zhigeng, "Dynamic Crowd Aggregation Simulation Using SIR Model Based Emotion Contagion," in *2017 International Conference on Virtual Reality and Visualization (ICVR)*, 2017, pp. 352–353.
- [4] B. Parkinson, "Interpersonal Emotion Transfer: Contagion and Social Appraisal," *Social and Personality Psychology Compass*, vol. 5, no. 7, pp. 428–439, 2011.
- [5] K. R. Scherer, A. Schorr, and T. Johnstone, *Appraisal processes in emotion: Theory, methods, research*. Oxford University Press, 2001.
- [6] K. R. Scherer, V. Shuman, J. J. R. Fontaine, and C. Soriano, "The GRID meets the Wheel: Assessing emotional feeling via self-report," in *Components of emotional meaning: A sourcebook*, New York, NY, US: Oxford University Press, 2013, pp. 281–298.